

Public Key Cryptography



UNIT III

PUBLIC KEY CRYPTOGRAPHY

MATHEMATICS OF ASYMMETRIC KEY CRYPTOGRAPHY: Primes –Primality Testing –Factorization – Euler’s totient function, Fermat’s andEuler’s Theorem – Chinese Remainder Theorem – Exponentiation andlogarithm – ASYMMETRIC KEY CIPHERS: RSA cryptosystem – Keydistribution – Key management – Diffie Hellman key exchange -ElGamal cryptosystem – Elliptic curve arithmetic-Elliptic curve cryptography.

Show moreShow less

UNIT III – PUBLIC KEY CRYPTOGRAPHY

1. Introduction to Public Key Cryptography

Public Key Cryptography (Asymmetric Key Cryptography) is a cryptographic system that uses two different keys:

1. **Public Key** → known to everyone
2. **Private Key** → kept secret by owner

A message encrypted using one key can only be decrypted using the other key.

Features

- Provides confidentiality
- Provides authentication
- Supports digital signatures
- Solves key distribution problem of symmetric encryption

Applications

- Secure communication
- HTTPS
- Digital signatures
- Email security
- Online banking
- Cryptocurrency

MATHEMATICS OF ASYMMETRIC KEY CRYPTOGRAPHY

2. Prime Numbers

A prime number is a positive integer greater than 1 having only two factors:

- 1
- itself

Examples

- 2, 3, 5, 7, 11, 13

Importance in Cryptography

Large prime numbers are the foundation of:

- RSA
- Diffie-Hellman
- ECC

Because factorization of large numbers is computationally difficult.

3. Primality Testing

Primality testing is the process of determining whether a number is prime or not.

Methods

(a) Trial Division Method

Check divisibility from 2 to $\sqrt{n}$.

Example:

Check whether 17 is prime.

$$\sqrt{17} \approx 4$$

Check divisibility by:

- 2 → not divisible
- 3 → not divisible
- 4 → not divisible

Hence 17 is prime.

(b) Fermat Primality Test

Based on Fermat's theorem:

If p is prime and a is not divisible by p ,

$$a^{p-1} \equiv 1 \pmod{p}$$

Used for testing large primes.

4. Factorization

Factorization means expressing a number as multiplication of prime numbers.

Example

$$60 = 2^2 \times 3 \times 5$$

Importance

RSA security depends on difficulty of factorizing very large numbers.

Example:

$$n = p \times q$$

Finding p and q from very large n is extremely difficult.

5. Euler's Totient Function

Euler's Totient Function is denoted by:

$$\phi(n)$$

It represents the number of integers less than n that are relatively prime to n .

Formula for Prime Number

If p is prime:

$$\phi(p) = p - 1$$

Example:

$$\phi(7) = 6$$

Formula for Product of Two Primes

If:

$$n = pq$$

then

$$\phi(n) = (p - 1)(q - 1)$$

Example:

$$n = 3 \times 5 = 15$$

$$\phi(15) = (3 - 1)(5 - 1) = 2 \times 4 = 8$$

6. Fermat's Theorem

If p is prime and $\gcd(a, p) = 1$,

$$a^{p-1} \equiv 1 \pmod{p}$$

Example

$$2^6 \equiv 1 \pmod{7}$$

$$64 \pmod{7} = 1$$

Hence verified.

7. Euler's Theorem

If $\gcd(a, n) = 1$,

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

This is generalized Fermat theorem.

Example

Take:

- $a = 3$
- $n = 10$

$$\phi(10) = 4$$

$$3^4 = 81$$

$$81 \bmod 10 = 1$$

Hence verified.

8. Chinese Remainder Theorem (CRT)

CRT helps solve simultaneous congruences.

Statement

If moduli are pairwise relatively prime:

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

Then there exists a unique solution modulo:

$$M = m_1 m_2$$

Example

Solve:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

Checking numbers satisfying first equation:

2, 5, 8, 11, 14...

11 satisfies second equation.

Hence:

$$x = 11$$

Applications

- RSA optimization
- Parallel computation
- Computer arithmetic

9. Exponentiation and Logarithm

Modular Exponentiation

Used for efficient computation of:

$$a^b \mod n$$

Example

$$3^4 \mod 5$$

$$81 \mod 5 = 1$$

Used heavily in:

- RSA
- Diffie-Hellman

Discrete Logarithm

If:

$$a^x \equiv b \pmod{n}$$

Finding x is called discrete logarithm problem.

Very hard computationally.

Used in:

- Diffie-Hellman
- ElGamal
- ECC

ASYMMETRIC KEY CIPHERS

10. RSA Cryptosystem

RSA is the most popular public key cryptosystem invented by:

- Rivest
- Shamir
- Adleman

RSA Algorithm

Step 1: Choose Two Prime Numbers

$$p = 3, q = 11$$

Step 2: Compute n

$$n = pq = 3 \times 11 = 33$$

Step 3: Compute Totient

$$\phi(n) = (3 - 1)(11 - 1) = 20$$

Step 4: Choose e

Choose e such that:

$$1 < e < 20$$

and

$$\gcd(e, 20) = 1$$

Take:

$$e = 7$$

Step 5: Find d

$$d \times e \equiv 1 \pmod{20}$$

$$7 \times 3 = 21 \equiv 1 \pmod{20}$$

So:

$$d = 3$$

Keys

Public Key

$$(e, n) = (7, 33)$$

Private Key

$$(d, n) = (3, 33)$$

Encryption

Message:

$$M = 2$$

Ciphertext:

$$C = M^e \mod n$$

$$C = 2^7 \mod 33$$

$$128 \mod 33 = 29$$

Encrypted message:

$$C = 29$$

Decryption

$$M = C^d \mod n$$

$$M = 29^3 \mod 33$$

$$M = 2$$

Original message recovered.

Advantages of RSA

- High security
- Supports digital signatures
- Secure key exchange

Disadvantages

- Slow for large data
 - Requires large keys
-

11. Key Distribution

Key distribution means sharing cryptographic keys securely between users.

Methods

1. Physical delivery
 2. Trusted third party
 3. Public key distribution
 4. Digital certificates
-

12. Key Management

Key management involves:

- Key generation
- Storage
- Distribution
- Updating
- Revocation
- Destruction

Good key management is essential for security.

13. Diffie-Hellman Key Exchange

Diffie-Hellman allows two users to establish a shared secret key over insecure channel.

Algorithm

Public Values

- Prime number p
- Primitive root g

Assume:

$$p = 23, g = 5$$

Alice Chooses Private Key

$$a = 6$$

Computes:

$$A = g^a \mod p$$

$$A = 5^6 \mod 23 = 8$$

Bob Chooses Private Key

$$b = 15$$

Computes:

$$B = 5^{15} \mod 23 = 19$$

Shared Secret

Alice Computes

$$K = B^a \mod p$$

$$K = 19^6 \mod 23 = 2$$

Bob Computes

$$K = A^b \mod p$$

$$K = 8^{15} \mod 23 = 2$$

Both get same key.

Advantages

- Secure key exchange
- No prior shared secret needed

Disadvantages

- Vulnerable to man-in-the-middle attack
-

14. ElGamal Cryptosystem

ElGamal is based on Diffie-Hellman and discrete logarithm problem.

Features

- Public key encryption
 - Digital signature support
 - Probabilistic encryption
-

Working Principle

Key Generation

- Choose prime p
- Choose generator g
- Select private key x
- Compute:

$$y = g^x \mod p$$

Public key:

$$(p, g, y)$$

Private key:

$$x$$

Encryption

Random number k chosen.

Ciphertext:

$$C_1 = g^k \mod p$$

$$C_2 = M \times y^k \mod p$$

Decryption

$$M = C_2 \times (C_1^x)^{-1} \mod p$$

Advantages

- More secure randomness
- Strong security

Disadvantages

- Ciphertext size larger
 - Slower than RSA
-

15. Elliptic Curve Arithmetic

Elliptic Curve Cryptography uses points on elliptic curves.

Elliptic Curve Equation

$$y^2 = x^3 + ax + b$$

Condition:

$$4a^3 + 27b^2 \neq 0$$

Operations on Elliptic Curves

(a) Point Addition

Adding two points on curve gives another point.

(b) Point Doubling

Adding a point to itself.

(c) Scalar Multiplication

$$kP = P + P + P + \dots$$

Repeated addition.

This operation is easy, but reverse operation is difficult.

16. Elliptic Curve Cryptography (ECC)

ECC is public key cryptography based on elliptic curves.

Security based on:

- Elliptic Curve Discrete Logarithm Problem (ECDLP)
-

Features of ECC

- Smaller key size
- Faster computation
- Less memory usage
- High security

Comparison with RSA

RSA	ECC
Large key required	Small key sufficient
Slower	Faster
More memory	Less memory
Heavy computation	Lightweight

Applications of ECC

- Mobile devices
- Smart cards
- Cryptocurrency
- SSL/TLS
- IoT security

Advantages of Public Key Cryptography

1. Secure communication
2. Digital signatures
3. Authentication
4. Easy key distribution
5. Supports e-commerce

Disadvantages

1. Computationally slower
2. Complex algorithms
3. Larger processing overhead

Conclusion

Public key cryptography is the foundation of modern secure communication systems. Mathematical concepts such as prime numbers, Euler theorem, factorization, and modular arithmetic form the basis of asymmetric algorithms like RSA, Diffie-Hellman, ElGamal, and ECC. These techniques provide confidentiality, authentication, integrity, and secure key exchange for modern digital communication systems.